LESSON 4.2

Relate Fractions, Decimals, and Percentages

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write equivalent fractions, decimals, and percents for the same value.

WHY THIS MATTERS You're designing the scoring system for a new arcade game. Players earn points as fractions of the total possible score. The leaderboard shows scores as percentages, but the game engine uses decimals. You need to convert between all three forms to build the display! A player scored 3/4 of the total possible points. Write that score as a decimal and as a percent.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Percent *Spanish: Porcentaje*	A way to compare a number to 100, shown with the % sign.	45% = 45 out of 100
Decimal *Spanish: Decimal*	A number with a dot, like 0.5, that shows a part less than one.	0.45 = 45 hundredths
Equivalent *Spanish: Equivalente*	Having the same value, just written a different way.	3/4 = 0.75 = 75%
Benchmark *Spanish: Referencia*	A familiar number you use to guess, like 50% or 1/2.	50% = one half
Discount *Spanish: Descuento*	Money taken off the first price to make it cheaper.	$40 shirt at 25% off: discount = $10
Hundredths *Spanish: Centésimos*	The second place after the decimal point; parts out of 100.	0.09 = 9 hundredths

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

20 students voted. 7 chose green. Write the green votes as a fraction, a decimal, and a percent.

1. Fraction: 7/20.
2. 20 × 5 = 100, so 7/20 = 35/100.
3. 35/100 = 0.35 = 35%.

Answer: 7/20 = 0.35 = 35%

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's change 1/5 into a fraction out of 100, then a decimal and a percent.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- _____ / _____ is the same value as _____ written as a decimal.
- To change a decimal into a percent I _____ .
- _____ % means _____ out of every hundred.
- I know these three forms are equivalent because _____ .

Word bank percent | decimal | fraction | equivalent | hundredths | denominator | numerator | simplest form | benchmark | scale | whole | part

WATCH OUT!

A common mistake

A common mistake in Relate Fractions, Decimals, and Percents is treating the numerator and denominator as the percent's digits

LESSON 4.2

Relate Fractions, Decimals, and Percentages

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort these values — which are greater than 50% and which are less than 50%?

Groups: Greater than 50% | Less than 50%

_____ 3/4

_____ 0.8

_____ 65%

_____ 7/10

_____ 1/3

_____ 0.25

_____ 40%

_____ 2/5

STEPS TO TRY

1. 50% is the same as 1/2 or 0.5 — that is your comparison line for every card.
2. Turn each card into a fraction out of 100 or a decimal, then ask: is it more or less than 50/100?
3. Try 3/4 first: 4 × 25 = 100, so multiply the top by 25 too — 3/4 = 75/100 = 0.75. Is 75/100 above or below 50/100? Sort it, then test the other cards the same way.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Convert 3/5 to a percent.

(A) 35%

(B) 53%

(C) 60%

(D) 65%

STEPS TO TRY

1. Percent means out of 100. What do you multiply the denominator 5 by to get 100?
2. 5 × 20 = 100, so multiply the numerator by that same 20 to build the equivalent fraction.
3. 3 × 20 = 60, so 3/5 = 60/100. A fraction out of 100 IS the percent.

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

3 **Circle the letter of the best answer. Show how you know.**

Convert 7/20 to a percent.

(A) 14%
(B) 20%
(C) 35%
(D) 70%

STEPS TO TRY

1. The denominator 20 is friendly: 20 × 5 = 100, and percents are always 'out of 100'.
2. Make an equivalent fraction with denominator 100 by multiplying top AND bottom by the same number.
3. Multiply 7/20 by 5/5: the bottom becomes 100. Whatever the top becomes is your percent.

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

4 **Circle the letter of the best answer. Show how you know.**

Which decimal is equivalent to 45%?

(A) 0.045
(B) 0.45
(C) 4.5
(D) 45.0

STEPS TO TRY

1. Percent means 'out of 100', so 45% is the fraction 45/100.
2. To turn a percent into a decimal, divide by 100 — the decimal point slides two places to the LEFT.
3. Start at 45. and move the decimal two places left: 45 → 4.5 → where does it land after the second hop?



0% **50%** **100%**

WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

EXPLAIN YOUR THINKING

What number did you multiply by to turn each denominator into 100, and how did that fraction out of 100 give you the percent?

SENTENCE STARTER I multiplied the top and bottom of ___ by ___ to get ___/100, so the percent is ___%.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help